# 23GE581 -Compiler Design Lab

## Ex. No : 1

**Date : 10-07-2026**

| **COMMON question for all groups** |
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| Write a Lex program to count the words, chars and lines in a file. |
| **GROUP 1** |
| Write a Lex program that find the longest string (without white spaces) in a file. |
| Write a Lex program to implement a lexical analyzer for a subset of the C language. It should recognize and categorize tokens into **Keywords**, **Identifiers**, **Constants**, **Operators**, and **Punctuation**. |
| Write a Lex program to distinguish between positive numbers, negative numbers, and fractions from a list of inputs. |
| Write a Lex program to separate tokens from an SQL select query with where and group by clauses. Operators such as like, = , > and < are included. |
| **GROUP 2** |
| Write a Lex program to eliminate all white space, comments (enclosed in /* and */) in document. |
| Write a Lex program to scan a text file and extract all valid email addresses and URLs, writing them into two separate files. |
| Write a Lex program that takes an HTML file as input and removes all HTML tags, leaving only the plain text. |
| Write a Lex program to read a text file and do the following<br>a. Add 1 to all integers.<br>b. Add 0.5 to all floating-point numbers. |
| **GROUP 3** |
| Write a Lex program to list the URLs in a html page. |
| Write a Lex program to recognize relational operators (<, <=, >, >=, ==, !=) in an input stream and print their corresponding meaning in words (e.g., print "Less than or equal to" when <= is encountered). |
| Write a Lex program that encrypts a text file by replacing each alphabetical character with the character three places down the alphabet (e.g., 'A' becomes 'D', 'z' becomes 'c'). Non-alphabetical characters should remain unchanged. |

Write a Lex program that counts the number of vowels and consonants in a given input text.

**GROUP 4**

Write a Lex program to divide the following C program into lexemes

```
 void main()
{
  int a, b=10;
  float c;
  c = a+b*10.3;
}
```

**Output**

```
<kw,void>  <id, main> ()
{
    <kw,int> <id,a> , <id,b> <op,=> 10i;
     <kw,float> <id,c>;
    <id,c>  <op,=> <id,a> <op,+> <id,b> <op,*> 10.3f;
}
```

Write a Lex program that reads a text file and capitalizes the first letter of every word that is exactly 5 characters long.

Write a Lex program to replace multiple consecutive spaces or tabs in a file with a single space.

Write a Lex program to check whether a given string is a valid C identifier or not (e.g., must start with a letter or underscore, followed by letters/digits).

**GROUP 5**

Write a Lex program to count the number of binary palindrome string in a given file and write them into a separate file.

Write a Lex program that find the longest string (without white spaces) in a file.

Write a Lex program to implement a lexical analyzer for a subset of the C language. It should recognize and categorize tokens into **Keywords**, **Identifiers**, **Constants**, **Operators**, and **Punctuation**.

Write a Lex program to distinguish between positive numbers, negative numbers, and fractions from a list of inputs.

**GROUP 6**

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| Write a Lex program to count the number of palindrome string over alphabet{a, b} in a given file and write them into a separate file. |
| Write a Lex program to eliminate all white space, comments (enclosed in /* and */) in document. |
| Write a Lex program to scan a text file and extract all valid email addresses and URLs, writing them into two separate files. |
| Write a Lex program that takes an HTML file as input and removes all HTML tags, leaving only the plain text. |
| **GROUP 7** |
| Write a Lex program to separate tokens from an SQL select query with where and group by clauses. Operators such as like, = , > and < are included. |
| Write a Lex program to list the URLs in a html page. |
| Write a Lex program to recognize relational operators (<, <=, >, >=, ==, !=) in an input stream and print their corresponding meaning in words (e.g., print "Less than or equal to" when <= is encountered). |
| Write a Lex program that encrypts a text file by replacing each alphabetical character with the character three places down the alphabet (e.g., 'A' becomes 'D', 'z' becomes 'c'). Non-alphabetical characters should remain unchanged. |
| **GROUP 8** |
| Write a Lex program to read a text file and do the following<br>a. Add 1 to all integers.<br>b. Add 0.5 to all floating-point numbers. |
| Write a Lex program to divide the following C program into lexemes<br>void main()<br>{<br>int a, b=10;<br>float c;<br>c = a+b*10.3;<br>}<br>**Output**<br><kw,void> <id, main> ()<br>{<br><kw,int> <id,a> , <id,b> <op,=> 10i;<br><kw,float> <id,c>; |

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|---|
| <id,c> <op,=> <id,a> <op,+> <id,b> <op,*> 10.3f;<br>} |
| Write a Lex program that reads a text file and capitalizes the first letter of every word that is exactly 5 characters long. |
| Write a Lex program to replace multiple consecutive spaces or tabs in a file with a single space. |
| **GROUP 9** |
| Write a Lex program that counts the number of vowels and consonants in a given input text. |
| Write a Lex program to count the number of binary palindrome string in a given file and write them into a separate file. |
| Write a Lex program that find the longest string (without white spaces) in a file. |
| Write a Lex program to implement a lexical analyzer for a subset of the C language. It should recognize and categorize tokens into **Keywords**, **Identifiers**, **Constants**, **Operators**, and **Punctuation**. |
| **GROUP 10** |
| Write a Lex program to check whether a given string is a valid C identifier or not (e.g., must start with a letter or underscore, followed by letters/digits). |
| Write a Lex program to count the number of palindrome string over alphabet{a, b} in a given file and write them into a separate file. |
| Write a Lex program to eliminate all white space, comments (enclosed in /* and */) in document. |
| Write a Lex program to scan a text file and extract all valid email addresses and URLs, writing them into two separate files. |
| **GROUP 11** |
| Write a Lex program to distinguish between positive numbers, negative numbers, and fractions from a list of inputs. |
| Write a Lex program to separate tokens from an SQL select query with where and group by clauses. Operators such as like, = , > and < are included. |
| Write a Lex program to recognize relational operators (<, <=, >, >=, ==, !=) in an input stream and print their corresponding meaning in words (e.g., print "Less than or equal to" when <= is encountered). |
| Write a Lex program to list the URLs in a html page. |

| GROUP 12 |
| --- |
| Write a Lex program that takes an HTML file as input and removes all HTML tags, leaving only the plain text. |
| Write a Lex program to read a text file and do the following<br>a. Add 1 to all integers.<br>b. Add 0.5 to all floating-point numbers. |
| Write a Lex program to divide the following C program into lexemes<br>void main()<br>{<br>int a, b=10;<br>float c;<br>c = a+b*10.3;<br>}<br>**Output**<br><kw,void> <id, main> ()<br>{<br><kw,int> <id,a> , <id,b> <op,=> 10i;<br><kw,float> <id,c>;<br><id,c> <op,=> <id,a> <op,+> <id,b> <op,*> 10.3f;<br>} |
| Write a Lex program that reads a text file and capitalizes the first letter of every word that is exactly 5 characters long. |
| **GROUP 13** |
| Write a Lex program that encrypts a text file by replacing each alphabetical character with the character three places down the alphabet (e.g., 'A' becomes 'D', 'z' becomes 'c'). Non-alphabetical characters should remain unchanged. |
| Write a Lex program that counts the number of vowels and consonants in a given input text. |
| Write a Lex program to count the number of binary palindrome string in a given file and write them into a separate file. |
| Write a Lex program that find the longest string (without white spaces) in a file. |
| **GROUP 14** |
| Write a Lex program to replace multiple consecutive spaces or tabs in a file with a single space. |

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| Write a Lex program to check whether a given string is a valid C identifier or not (e.g., must start with a letter or underscore, followed by letters/digits). |
| Write a Lex program to count the number of palindrome string over alphabet{a, b} in a given file and write them into a separate file. |
| Write a Lex program to eliminate all white space, comments (enclosed in /* and */) in document. |